HSC Mathematics Advanced — Practice Paper

Topic: Sequences and Series (Arithmetic & Geometric)

Questions adapted from NSW school Trial HSC Examinations 2025–2026 (MA12-4) • Each question is followed by a NESA-style sample answer and marking criteria • Total: 79 questions, 208 marks

Difficulty labels: EASY | MODERATE | DIFFICULT

NESA-approved calculators may be used. Key formulae are summarised on page 20.

Question 1 (1 mark) — EASY

Adapted from Blacktown Boys High School 2025 Trial HSC

Which of the following is an arithmetic sequence?

(A) $6,\ 24,\ 96,\ \ldots$

(B) $9,\ 14,\ 20,\ \ldots$

(C) $27,\ 34,\ 41,\ \ldots$

(D) $48,\ -24,\ 12,\ \ldots$

Sample answer

Arithmetic $\Leftrightarrow$ constant difference between consecutive terms

$27,\ 34,\ 41$: $34-27=41-34=7$ ✓ (A and D are geometric; B has differences 5, 6)

Marking criteria

Criteria	Marks
• Correct answer: **C**	1

Question 2 (1 mark) — EASY

Adapted from Blacktown Boys High School 2026 Trial HSC

The first three terms of an arithmetic sequence are 5, 9 and 13. What is the 18th term of the sequence?

(A) 81

(B) 77

(C) 73

(D) 69

Sample answer

$a=5,\ d=4$

$T_{18}=5+17\times4=73$

Marking criteria

Criteria	Marks
• Correct answer: **C**	1

Question 3 (1 mark) — EASY

Adapted from North Sydney Boys High School 2026 Trial HSC

What is the 20th term of the arithmetic sequence $-7,\ -3,\ 1,\ 5,\ \ldots$?

(A) 65

(B) 69

(C) 73

(D) 77

Sample answer

$a=-7,\ d=4$

$T_{20}=-7+19\times4=69$

Marking criteria

Criteria	Marks
• Correct answer: **B**	1

Question 4 (1 mark) — EASY

Adapted from Manly Selective Campus 2026 Trial HSC

The numbers $5,\ x,\ 13,\ 17,\ \ldots$ form an arithmetic sequence. What is the sum of the first four terms?

(A) 35

(B) 39

(C) 42

(D) 44

Sample answer

$d=17-13=4 \Rightarrow x=9$

$S_4=5+9+13+17=44$

Marking criteria

Criteria	Marks
• Correct answer: **D**	1

Question 5 (1 mark) — EASY

Adapted from Fort Street High School 2026 Trial HSC

In an arithmetic series, the first term is -2 and the third term is 8. What is the 10th term?

(A) 38

(B) 41

(C) 43

(D) 45

Sample answer

$a+2d=8 \Rightarrow 2d=10,\ d=5$

$T_{10}=-2+9\times5=43$

Marking criteria

Criteria	Marks
• Correct answer: **C**	1

Question 6 (1 mark) **EASY**

Adapted from Caringbah High School 2026 Trial HSC

In an arithmetic series, the first term is -2 and the third term is 8. What is the sum of the first 15 terms?

(A) 395 (B) 450

(C) 495 (D) 540

Sample answer

$d=5$

$S_{15}=\frac{15}{2}[2(-2)+14\times5]=\frac{15}{2}\times66=495$

Marking criteria

Criteria	Marks
• Correct answer: **C**	1

Question 7 (1 mark) **EASY**

Adapted from Hunters Hill High School 2026 Trial HSC

The first three terms of an arithmetic sequence are $1,\ 3,\ 5,\ \ldots$. Which of the following gives the sum of the first n terms, S_n?

(A) $S_n=n^2$ (B) $S_n=2n$

(C) $S_n=2n-1$ (D) $S_n=2n+1$

Sample answer

$S_n=\frac{n}{2}[2(1)+(n-1)2]=\frac{n}{2}\times2n=n^2$

Marking criteria

Criteria	Marks
• Correct answer: **A**	1

Question 8 (1 mark) **EASY**

Adapted from Hunters Hill High School 2025 Trial HSC

Find the limiting sum of the series $2\pi+\pi+\frac{\pi}{2}+\cdots$

(A) $\frac{\pi}{2}$ (B) π

(C) 2π (D) 4π

Sample answer

$a=2\pi,\ r=\frac{1}{2}$

$S_\infty=\frac{2\pi}{1-\frac{1}{2}}=4\pi$

Marking criteria

Criteria	Marks
• Correct answer: **D**	1

Question 9 (1 mark) **EASY**

Adapted from St George Girls High School 2025 Trial HSC

Consider the series $1+1+1+1+\cdots$. Which statement could NOT be made about the series?

(A) It is an arithmetic progression with common difference 0 (B) It is a geometric progression with common ratio 1

(C) It has a limiting sum of $2n$, where n is the number of terms (D) Its terms satisfy $T_n=T_{n-1}$, where $T_1=1$

Sample answer

$d=0$ (A true), $r=1$ (B true), $T_n=T_{n-1}$ (D true)

A limiting sum requires $|r|<1$; with $r=1$ no limiting sum exists, so C is false

Marking criteria

Criteria	Marks
• Correct answer: **C**	1

Question 10 (1 mark) **EASY**

Adapted from Blacktown Boys High School 2025 Trial HSC

Find the 100th term of the sequence $10,\ 6\frac{1}{2},\ 3,\ \ldots$

(A) $356\frac{1}{2}$ (B) $-336\frac{1}{2}$

(C) -333 (D) 353

Sample answer

$a=10,\ d=-3\frac{1}{2}$

$T_{100}=10+99\times\left(-3\frac{1}{2}\right)=10-346\frac{1}{2}=-336\frac{1}{2}$

Marking criteria

Criteria	Marks
• Correct answer: **B**	1

Question 11 (1 mark) MODERATE

Adapted from Killara High School 2025 Trial HSC

Consider the geometric sequence $81,\ a,\ b,\ 3$. Find the value of $b-a$.

(A) -18 (B) 18

(C) -26 (D) 26

Sample answer

$81r^3=3 \Rightarrow r^3=\frac{1}{27},\ r=\frac{1}{3}$

$a=27,\ b=9 \Rightarrow b-a=-18$

Marking criteria

Criteria	Marks
• Correct answer: **A**	1

Question 12 (1 mark) MODERATE

Adapted from Hurlstone Agricultural High School 2026 Trial HSC

If $x^2,\ 4x+3,\ 25$ are in a geometric sequence, what is the value of x?

(A) $-\frac{1}{3}$ (B) -3

(C) $\frac{1}{3}$ or -3 (D) $-\frac{1}{3}$ or 3

Sample answer

$(4x+3)^2=25x^2 \Rightarrow 4x+3=\pm 5x$

$4x+3=5x \Rightarrow x=3;\ 4x+3=-5x \Rightarrow x=-\frac{1}{3}$

Marking criteria

Criteria	Marks
• Correct answer: **D**	1

Question 13 (1 mark) MODERATE

Adapted from Killara High School 2026 Trial HSC

A geometric series is given by $4+2x+x^2+\frac{x^3}{2}+\ldots$. If the limiting sum of the series is 10, what is the value of x?

(A) $\frac{1}{2}$ (B) $\frac{3}{5}$

(C) $\frac{4}{5}$ (D) $\frac{6}{5}$

Sample answer

$a=4,\ r=\frac{2x}{4}=\frac{x}{2}$

$\frac{4}{1-\frac{x}{2}}=10 \Rightarrow 1-\frac{x}{2}=\frac{2}{5} \Rightarrow x=\frac{6}{5}$

Marking criteria

Criteria	Marks
• Correct answer: **D**	1

Question 14 (1 mark) MODERATE

Adapted from Blacktown Boys High School 2025 Trial HSC

Which expression is a term of the geometric series $2x-4x^2+8x^3-16x^4+\cdots$?

(A) $-2048x^{11}$ (B) $2048x^{11}$

(C) $-2048x^{10}$ (D) $2048x^{10}$

Sample answer

$a=2x,\ r=-2x:\ T_n=2x(-2x)^{n-1}=(-1)^{n-1}2^n x^n$

Odd n gives a positive coefficient: $T_{11}=2^{11}x^{11}=2048x^{11}$

Marking criteria

Criteria	Marks
• Correct answer: **B**	1

Question 15 (1 mark) MODERATE

Adapted from North Sydney Boys High School 2025 Trial HSC

For an arithmetic sequence, $T_4+T_5=24$ and $S_6=48$. What is the common difference d?

(A) 1 (B) 2

(C) 4 (D) 8

Sample answer

$T_4+T_5=2a+7d=24$ ... (1)

$S_6=3(2a+5d)=48 \Rightarrow 2a+5d=16$ ... (2)

(1) − (2): $2d=8,\ d=4$

Marking criteria

Criteria	Marks
• Correct answer: **C**	1

Question 16 (1 mark) MODERATE

Adapted from Penrith High School 2025 Trial HSC

The formula for the nth term of a series is $T_n = 400n - 2^n$. What is the sum of the first 10 terms?

(A) 2 976

(B) 407 154

(C) $-135\,718$

(D) 19 954

Sample answer

$\sum_{n=1}^{10} 400n = 400 \times \frac{10 \times 11}{2} = 22\,000$

$\sum_{n=1}^{10} 2^n = \frac{2(2^{10}-1)}{2-1} = 2046;\ S_{10} = 22\,000 - 2046 = 19\,954$

Marking criteria

Criteria	Marks
• Correct answer: **D**	1

Question 17 (1 mark) MODERATE

Adapted from Girraween High School 2025 Trial HSC

The first term of a geometric series is 4 and the third term is 9. The common ratio is negative. The 40th term of the series is

(A) $4\left(\frac{3}{2}\right)^{39}$

(B) $-4\left(\frac{3}{2}\right)^{39}$

(C) $4\left(\frac{3}{2}\right)^{40}$

(D) $-4\left(\frac{3}{2}\right)^{40}$

Sample answer

$4r^2 = 9 \Rightarrow r = -\frac{3}{2}$ (negative)

$T_{40} = 4\left(-\frac{3}{2}\right)^{39} = -4\left(\frac{3}{2}\right)^{39}$

Marking criteria

Criteria	Marks
• Correct answer: **B**	1

Question 18 (1 mark) MODERATE

Adapted from Manly Selective Campus 2025 Trial HSC

Let $0 < q < p$. If p and q are the 1st and 2nd terms of a geometric progression, what is the limiting sum?

(A) $\frac{p}{1-q}$

(B) $\frac{pq}{q-p}$

(C) $\frac{p^2}{p+q}$

(D) $\frac{p^2}{p-q}$

Sample answer

$a = p,\ r = \frac{q}{p}$ with $0 < r < 1$

$S_\infty = \frac{p}{1-\frac{q}{p}} = \frac{p^2}{p-q}$

Marking criteria

Criteria	Marks
• Correct answer: **D**	1

Question 19 (1 mark) DIFFICULT

Adapted from Fort Street High School 2026 Trial HSC

The limiting sum of a geometric series is $\frac{10}{9}$ and the sum of its first three terms is $\frac{26}{25}$. What is the common ratio of this series?

(A) 1.39

(B) 1.25

(C) 0.40

(D) 0.25

Sample answer

$S_3 = \frac{a(1-r^3)}{1-r} = S_\infty(1-r^3) \Rightarrow \frac{26}{25} = \frac{10}{9}(1-r^3)$

$1 - r^3 = \frac{26}{25} \times \frac{9}{10} = 0.936 \Rightarrow r^3 = 0.064,\ r = 0.4$

Marking criteria

Criteria	Marks
• Correct answer: **C**	1

Question 20 (2 marks) EASY

Adapted from Barker College 2025 Trial HSC

Find the limiting sum of the series $96 + 72 + 54 + \cdots$ **2**

Sample answer

$r = \frac{72}{96} = \frac{3}{4}$, $|r| < 1$ so the limiting sum exists

$S_\infty = \frac{96}{1-\frac{3}{4}} = 384$

Marking criteria

Criteria	Marks
• Correct answer	2
• Finds the common ratio, or equivalent merit	1

Question 21 (1 mark) DIFFICULT

Adapted from Manly Selective Campus 2025 Trial HSC

In a geometric sequence $T_3 = -45$, $S_5 = -305$ and $S_6 = 910$. What is the 9th term of the sequence?

(A) $-32\,805$ (B) $-10\,935$

(C) $10\,935$ (D) $32\,805$

Sample answer

$T_6 = S_6 - S_5 = 910 - (-305) = 1215$

$\frac{T_6}{T_3} = r^3 = \frac{1215}{-45} = -27 \Rightarrow r = -3$

$a = \frac{T_3}{r^2} = \frac{-45}{9} = -5;\ T_9 = -5(-3)^8 = -32\,805$

Marking criteria

Criteria	Marks
• Correct answer: **A**	1

Question 22 (1 mark) DIFFICULT

Adapted from Abbotsleigh 2025 Trial HSC

What is the limiting sum of the geometric series $\log_e 2 - (\log_e 2)^2 + (\log_e 2)^3 - (\log_e 2)^4 + \cdots$?

(A) $\log_{\frac{e}{2}} 2$ (B) $\log_{\frac{2}{e}} 2$

(C) $\log_{2+e} 2$ (D) $\log_{2e} 2$

Sample answer

$a = \ln 2,\ r = -\ln 2$ ($|r| < 1$ since $\ln 2 \approx 0.69$)

$S_\infty = \frac{\ln 2}{1 + \ln 2} = \frac{\ln 2}{\ln e + \ln 2} = \frac{\ln 2}{\ln(2e)} = \log_{2e} 2$

Marking criteria

Criteria	Marks
• Correct answer: **D**	1

Question 23 (3 marks) EASY

Adapted from Blacktown Boys High School 2025 Trial HSC

Consider the geometric series $18 + 6 + 2 + \cdots$

(a) Find the common ratio. **1**

(b) Find the 8th term. **1**

(c) Find the limiting sum of this series. **1**

Sample answer

(a)

$r = \frac{6}{18} = \frac{1}{3}$

(b)

$T_8 = 18\left(\frac{1}{3}\right)^7 = \frac{18}{2187} = \frac{2}{243}$

(c)

$S_\infty = \frac{18}{1 - \frac{1}{3}} = 27$

Marking criteria

Criteria	Marks
• Correct answer	1

Criteria	Marks
• Correct answer	1

Criteria	Marks
• Correct answer	1

Question 24 (2 marks) EASY

Adapted from Blacktown Boys High School 2025 Trial HSC

In an arithmetic sequence, the sixth term is 37 and the nineteenth term is 193. Find the value of the first term, a, and the common difference, d. **2**

Sample answer

$a + 5d = 37$ and $a + 18d = 193$

Subtracting: $13d = 156 \Rightarrow d = 12;\ a = 37 - 60 = -23$

Marking criteria

Criteria	Marks
• Correct values	2
• Forms two correct equations, or equivalent merit	1

Question 25 (2 marks) EASY

Adapted from Blacktown Boys High School 2026 Trial HSC

Find the sum of the terms in the arithmetic series $2026 + 2011 + 1996 + \cdots + 271$. **2**

Sample answer

$a = 2026,\ d = -15,\ l = 271:\ 271 = 2026 - 15(n-1) \Rightarrow n - 1 = 117,\ n = 118$

$S_{118} = \frac{118}{2}(2026 + 271) = 59 \times 2297 = 135\,523$

Marking criteria

Criteria	Marks
• Correct answer	2
• Finds the number of terms, or equivalent merit	1

Question 26 (2 marks) EASY

Adapted from Blacktown Boys High School 2026 Trial HSC

In a geometric sequence, the 10th term is 768 and the 13th term is 6144. Find the first term and the common ratio. **2**

Sample answer

$\frac{ar^{12}}{ar^9} = r^3 = \frac{6144}{768} = 8 \Rightarrow r = 2$

$a \times 2^9 = 768 \Rightarrow a = \frac{768}{512} = 1.5$

Marking criteria

Criteria	Marks
• Correct values	2
• Finds the common ratio, or equivalent merit	1

Question 27 (3 marks) EASY

Adapted from Baulkham Hills High School 2026 Trial HSC

An arithmetic sequence from start to end is $51,\ 44,\ 37,\ \ldots,\ -47$.

(a) Write a simplified expression for the nth term of this sequence. **1**

(b) How many terms are there in the sequence? **1**

(c) Find the sum of this series. **1**

Sample answer

(a)

$T_n = 51 + (n-1)(-7) = 58 - 7n$

(b)

$58 - 7n = -47 \Rightarrow 7n = 105,\ n = 15$

(c)

$S_{15} = \frac{15}{2}(51 + (-47)) = \frac{15}{2} \times 4 = 30$

Marking criteria

Criteria	Marks
• Correct expression	1

Criteria	Marks
• Correct answer	1

Criteria	Marks
• Correct answer	1

Question 28 (2 marks) EASY

Adapted from Hornsby Girls High School 2025 Trial HSC

The first three terms of an arithmetic sequence are $x+6,\ 2x-5$ and $4x-11$. Find the value of x. **2**

Sample answer

$T_2 - T_1 = T_3 - T_2:\ (2x-5) - (x+6) = (4x-11) - (2x-5)$

$x - 11 = 2x - 6 \Rightarrow x = -5$

Marking criteria

Criteria	Marks
• Correct answer	2
• Equates the two common differences, or equivalent merit	1

Question 29 (4 marks) EASY

Adapted from Hunters Hill High School 2025 Trial HSC

The tenth term of an arithmetic series is 29 and the fifteenth term is 44.

(a) Find the value of the common difference and the value of the first term. **2**

(b) Find the sum of the first 75 terms. **2**

Sample answer

(a)

$a + 9d = 29,\ a + 14d = 44 \Rightarrow 5d = 15,\ d = 3$

$a = 29 - 27 = 2$

(b)

$S_{75} = \frac{75}{2}[2(2) + 74(3)] = \frac{75}{2} \times 226 = 8475$

Marking criteria

Criteria	Marks
• Correct values	2
• Forms two correct equations, or equivalent merit	1

Criteria	Marks
• Correct answer	2
• Correct substitution into the sum formula, or equivalent merit	1

Question 30 (2 marks) EASY

Adapted from Hunters Hill High School 2026 Trial HSC

The numbers $1280,\ p,\ q,\ 20$ form a geometric sequence. Find the values of p and q. **2**

Sample answer

$1280r^3 = 20 \Rightarrow r^3 = \frac{1}{64},\ r = \frac{1}{4}$

$p = 1280 \times \frac{1}{4} = 320, \quad q = 80$

Marking criteria

Criteria	Marks
• Correct values	2
• Finds the common ratio, or equivalent merit	1

Question 31 (3 marks) EASY

Adapted from Killara High School 2025 Trial HSC

Find the sum of the terms in the geometric series $2+6+18+54+\cdots+118\,098$. **3**

Sample answer

$a=2,\ r=3:\ T_n=2\times3^{n-1}=118\,098 \Rightarrow 3^{n-1}=59\,049=3^{10}$, so $n=11$

$S_{11}=\dfrac{2(3^{11}-1)}{3-1}=3^{11}-1=177\,146$

Marking criteria

Criteria	Marks
• Correct answer	3
• Identifies that the series has 11 terms, or equivalent merit	2
• Identifies a and r, or equivalent merit	1

Question 32 (2 marks) EASY

Adapted from Knox Grammar School 2026 Trial HSC

Find the sum of the arithmetic series $180+165+150+\cdots-15-30$. **2**

Sample answer

$d=-15:\ -30=180-15(n-1) \Rightarrow n-1=14,\ n=15$

$S_{15}=\dfrac{15}{2}(180-30)=1125$

Marking criteria

Criteria	Marks
• Correct answer	2
• Finds the number of terms, or equivalent merit	1

Question 33 (2 marks) EASY

Adapted from Manly Selective Campus 2025 Trial HSC

A sequence is defined recursively by $T_1=256$ and $T_n=\dfrac{T_{n-1}}{4}$.

(a) List the first four terms of the sequence. **1**

(b) Find the limiting sum of the series formed by the sequence, if it exists. If it does not exist, explain why. **1**

Sample answer

(a)

$256,\ 64,\ 16,\ 4$

(b)

Geometric with $r=\dfrac{1}{4}$, $|r|<1$ so it exists: $S_\infty=\dfrac{256}{1-\frac{1}{4}}=\dfrac{1024}{3}$

Marking criteria

Criteria	Marks
• Correct terms	1

Criteria	Marks
• Correct answer with justification	1

Question 34 (2 marks) EASY

Adapted from Manly Selective Campus 2026 Trial HSC

In an arithmetic sequence, the 6th term is 36 and the 9th term is 28. Find the first term. **2**

Sample answer

$a+5d=36,\ a+8d=28 \Rightarrow 3d=-8,\ d=-\dfrac{8}{3}$

$a=36-5\left(-\dfrac{8}{3}\right)=36+\dfrac{40}{3}=\dfrac{148}{3}$

Marking criteria

Criteria	Marks
• Correct answer	2
• Finds the common difference, or equivalent merit	1

Question 35 (3 marks) EASY

Adapted from Penrith High School 2025 Trial HSC

Find the sum of the terms in the arithmetic series $71+64+57+\cdots+(-48)$. **3**

Sample answer

$d=-7:\ -48=71-7(n-1) \Rightarrow n-1=17,\ n=18$

$S_{18}=\dfrac{18}{2}(71-48)=9\times23=207$

Marking criteria

Criteria	Marks
• Correct answer	3
• Finds the number of terms, or equivalent merit	2
• Identifies d = -7, or equivalent merit	1

Question 36 (2 marks) EASY

Adapted from Roseville College 2026 Trial HSC

An arithmetic series has a common difference of 6 and its 20th term is 130. Find the sum of the first twenty terms of the series. **2**

Sample answer

$a+19\times6=130 \Rightarrow a=16$

$S_{20}=\dfrac{20}{2}(16+130)=1460$

Marking criteria

Criteria	Marks
• Correct answer	2
• Finds the first term, or equivalent merit	1

Question 37 (3 marks) — EASY

Adapted from St George Girls High School 2025 Trial HSC

The nth term of an arithmetic sequence is given by $T_n = 5n - 3$.

(a) Write down the first 3 terms of the sequence. **1**

(b) Find the sum of the first 50 terms. **2**

Sample answer

(a)

$T_1 = 2,\ T_2 = 7,\ T_3 = 12$

(b)

$T_{50} = 247;\ S_{50} = \frac{50}{2}(2 + 247) = 25 \times 249 = 6225$

Marking criteria

Criteria	Marks
• Correct terms	1

Criteria	Marks
• Correct answer	2
• Finds T_{50} or substitutes correctly, or equivalent merit	1

Question 38 (2 marks) — EASY

Adapted from St George Girls High School 2025 Trial HSC

Determine the last term under 100 in the sequence $-11,\ -4,\ 3,\ 10,\ 17,\ \ldots$ **2**

Sample answer

$T_n = -11 + 7(n-1) = 7n - 18 < 100 \Rightarrow n < 16.86$, so $n = 16$

$T_{16} = 7(16) - 18 = 94$

Marking criteria

Criteria	Marks
• Correct answer	2
• Finds T_n and sets up an inequality, or equivalent merit	1

Question 39 (2 marks) — EASY

Adapted from Sydney Girls High School 2025 Trial HSC

Consider the geometric series $7 + 14e^{-2} + 28e^{-4} + 56e^{-6} + \cdots$. Justify whether this series has a limiting sum or not. **2**

Sample answer

$r = \frac{14e^{-2}}{7} = 2e^{-2} = \frac{2}{e^2} \approx 0.27$

Since $|r| < 1$, the series has a limiting sum $\left(S_\infty = \frac{7}{1 - 2e^{-2}} \approx 9.6\right)$

Marking criteria

Criteria	Marks
• Correct conclusion with justification	2
• Finds the common ratio, or equivalent merit	1

Question 40 (2 marks) — EASY

Adapted from Sydney Girls High School 2026 Trial HSC

The first term of an arithmetic series is -7 and the sum of the first 20 terms is 430. Find the common difference. **2**

Sample answer

$S_{20} = \frac{20}{2}[2(-7) + 19d] = 430 \Rightarrow -14 + 19d = 43$

$19d = 57 \Rightarrow d = 3$

Marking criteria

Criteria	Marks
• Correct answer	2
• Correct substitution into the sum formula, or equivalent merit	1

Question 41 (2 marks) — EASY

Adapted from Sydney Grammar School 2025 Trial HSC

Find the value of the arithmetic series $1 + 5 + 9 + \cdots + 2025$. **2**

Sample answer

$2025 = 1 + 4(n-1) \Rightarrow n = 507$

$S_{507} = \frac{507}{2}(1 + 2025) = 513\,591$

Marking criteria

Criteria	Marks
• Correct answer	2
• Finds the number of terms, or equivalent merit	1

Question 42 (2 marks) — EASY

Adapted from Sydney Technical High School 2026 Trial HSC

Find the first term of the series $-12 - 8 - 4 + \cdots$ that is greater than 150. **2**

Sample answer

$T_n = -12 + 4(n-1) = 4n - 16 > 150 \Rightarrow n > 41.5$, so $n = 42$

$T_{42} = 4(42) - 16 = 152$

Marking criteria

Criteria	Marks
• Correct answer	2
• Sets up a correct inequality for n, or equivalent merit	1

Question 43 (3 marks) — EASY

Adapted from Sydney Girls High School 2025 Trial HSC

Consider the arithmetic sequence $-7,\ 2,\ 11,\ 20,\ \ldots,\ 713$.

(a) How many terms are in the sequence? **1**

(b) Find the sum of all the terms. **2**

Sample answer

(a)

$d=9:\ 713=-7+9(n-1) \Rightarrow n-1=80,\ n=81$

(b)

$S_{81}=\frac{81}{2}(-7+713)=\frac{81}{2}\times 706=28\,593$

Marking criteria

Criteria	Marks
• Correct answer	1

Criteria	Marks
• Correct answer	2
• Correct substitution into the sum formula, or equivalent merit	1

Question 44 (2 marks) — EASY

Adapted from North Sydney Boys High School 2026 Trial HSC

Find the sum of the arithmetic series $2+5+8+\cdots+53$. **2**

Sample answer

$53=2+3(n-1) \Rightarrow n=18$

$S_{18}=\frac{18}{2}(2+53)=495$

Marking criteria

Criteria	Marks
• Correct answer	2
• Finds the number of terms, or equivalent merit	1

Question 45 (3 marks) — EASY

Adapted from St George Girls High School 2025 Trial HSC

Find the sum of the geometric series $2+6+18+\cdots+39\,366$. **3**

Sample answer

$2\times 3^{n-1}=39\,366 \Rightarrow 3^{n-1}=19\,683=3^9$, so $n=10$

$S_{10}=\frac{2(3^{10}-1)}{3-1}=3^{10}-1=59\,048$

Marking criteria

Criteria	Marks
• Correct answer	3
• Finds the number of terms, or equivalent merit	2
• Identifies $r = 3$ and sets $T_n = 39\,366$, or equivalent merit	1

Question 46 (4 marks) — MODERATE

Adapted from Barker College 2025 Trial HSC

An arithmetic series is given by $118+105+92+79+\cdots$

(a) Find the 10th term of the series. **1**

(b) Find the minimum number of terms needed for the sum of the series to be negative. **3**

Sample answer

(a)

$a=118,\ d=-13:\ T_{10}=118+9(-13)=1$

(b)

$S_n=\frac{n}{2}[236-13(n-1)]=\frac{n}{2}(249-13n)<0$

Since $n>0$: $249-13n<0 \Rightarrow n>19.15$

Minimum $n=20$ (check: $S_{19}=19,\ S_{20}=-110$)

Marking criteria

Criteria	Marks
• Correct answer	1

Criteria	Marks
• Correct answer	3
• Solves the inequality for n, or equivalent merit	2
• Writes $S_n < 0$ using the sum formula, or equivalent merit	1

Question 47 (2 marks) — MODERATE

Adapted from Blacktown Boys High School 2026 Trial HSC

Evaluate the limiting sum of the series $\frac{5}{6}+\frac{5}{27}+\frac{10}{243}+\cdots$ **2**

Sample answer

$r=\frac{5}{27}\div\frac{5}{6}=\frac{2}{9}$ (check: $\frac{10}{243}\div\frac{5}{27}=\frac{2}{9}$)

$S_\infty=\frac{5/6}{1-\frac{2}{9}}=\frac{5}{6}\times\frac{9}{7}=\frac{15}{14}$

Marking criteria

Criteria	Marks
• Correct answer	2
• Finds the common ratio, or equivalent merit	1

Question 48 (5 marks) — MODERATE

Adapted from Caringbah High School 2025 / Hills Grammar 2025 Trial HSC

Finn accidentally puts a hole in the bottom of his rainwater tank, which contains 10 200 litres of water. In the first hour 200 litres leak out, and in the second hour a further 250 litres leak out. Finn wonders whether the amount leaking each hour forms an arithmetic or a geometric sequence.

(a) If the amount of water that leaks out each hour forms an arithmetic sequence, how many hours will it take until the tank is empty? **3**

(b) If the amount of water that leaks out each hour forms a geometric sequence, show that eight hours after the leak begins, over 60% of the water remains in the tank. **2**

Sample answer

(a)

$a = 200,\ d = 50:\ \frac{n}{2}[400 + 50(n-1)] = 10\,200$

$n(350 + 50n) = 20\,400 \Rightarrow n^2 + 7n - 408 = 0 \Rightarrow (n+24)(n-17) = 0$

$n = 17$ hours

(b)

$r = \frac{250}{200} = 1.25:\ S_8 = \frac{200(1.25^8 - 1)}{0.25} = 3968.37$ L

Remaining $= 10\,200 - 3968.37 = 6231.63$ L $= 61.1\% > 60\%$ ✓

Marking criteria

Criteria	Marks
• Correct answer	3
• Obtains a correct quadratic equation in n, or equivalent merit	2
• Sets $S_n = 10\,200$ with $a = 200$, $d = 50$, or equivalent merit	1

Criteria	Marks
• Correct proof	2
• Finds the total leaked after 8 hours, or equivalent merit	1

Question 49 (3 marks) — MODERATE

Adapted from Hornsby Girls High School 2025 Trial HSC

The sum of the first n terms of an arithmetic sequence is $S_n = 3n^2 - 4n$, where $n > 1$. Find the nth term, T_n, and state the common difference. **3**

Sample answer

$T_n = S_n - S_{n-1} = 3n^2 - 4n - [3(n-1)^2 - 4(n-1)]$

$= 3n^2 - 4n - 3n^2 + 6n - 3 + 4n - 4 = 6n - 7$

$d = T_2 - T_1 = 5 - (-1) = 6$ (coefficient of n)

Marking criteria

Criteria	Marks
• Correct T_n and d	3
• Correct expression for T_n, or equivalent merit	2
• Uses $T_n = S_n - S_{n-1}$, or equivalent merit	1

Question 50 (5 marks) — MODERATE

Adapted from Hornsby Girls High School 2026 Trial HSC

A musical instrument has many strings. The difference between the lengths of adjacent strings is constant, so the lengths form an arithmetic sequence. The shortest string is 30 cm and the longest is 50 cm. The sum of the lengths of all the strings is 1240 cm.

(a) Find the number of strings. **2**

(b) Find the difference in length between adjacent strings. **2**

(c) What is the length of the tenth string? **1**

Sample answer

(a)

$\frac{n}{2}(30 + 50) = 1240 \Rightarrow 40n = 1240,\ n = 31$

(b)

$30 + 30d = 50 \Rightarrow d = \frac{20}{30} = \frac{2}{3}$ cm

(c)

$T_{10} = 30 + 9 \times \frac{2}{3} = 36$ cm

Marking criteria

Criteria	Marks
• Correct answer	2
• Uses $S_n = n(a + l)/2$, or equivalent merit	1

Criteria	Marks
• Correct answer	2
• Uses $T_{31} = 50$, or equivalent merit	1

Criteria	Marks
• Correct answer	1

Question 51 (3 marks) MODERATE

Adapted from Hurlstone Agricultural High School 2026 Trial HSC

In an arithmetic series, the 1st term is -12, the 17th term is 12 and the last term is 45. Find the sum of all the terms in the series. **3**

Sample answer

$-12+16d=12 \Rightarrow d=\frac{3}{2}$

$45=-12+\frac{3}{2}(n-1) \Rightarrow n-1=38,\ n=39$

$S_{39}=\frac{39}{2}(-12+45)=643.5$

Marking criteria

Criteria	Marks
• Correct answer	3
• Finds the number of terms, or equivalent merit	2
• Finds the common difference, or equivalent merit	1

Question 52 (5 marks) MODERATE

Adapted from James Ruse Agricultural High School 2025 Trial HSC

A venetian blind consists of twenty slats, each 2.5 mm thick. When the blind is down, the gap between the top slat and the top of the blind is 27 mm and the gap between adjacent slats is also 27 mm. When the blind is up, all the slats are stacked at the top with no gaps.

(a) Show that when the blind is raised, the bottom slat rises 540 mm. **2**

(b) How far does the 19th slat rise? **1**

(c) Explain briefly why the distances the slats rise form an arithmetic sequence. **1**

(d) Find the sum of all the distances that the slats rise when the blind is raised. **1**

Sample answer

(a)

Down: top of 20th slat is $20\times27+19\times2.5=587.5$ mm below the top

Up: top of 20th slat is $19\times2.5=47.5$ mm below the top. Rise $=587.5-47.5=540$ mm

Marking criteria

Criteria	Marks
• Correct proof	2
• Finds the position of the bottom slat when down or up, or equivalent merit	1

(b)

Slat k loses k gaps of 27 mm: 19th slat rises $19\times27=513$ mm

Criteria	Marks
• Correct answer	1

(c)

Each successive slat rises 27 mm more than the one above it (constant difference 27)

Criteria	Marks
• Correct explanation	1

(d)

$S_{20}=\frac{20}{2}(27+540)=5670$ mm

Criteria	Marks
• Correct answer	1

Question 53 (4 marks) MODERATE

Adapted from James Ruse Agricultural High School 2026 Trial HSC

A wildlife camera records video clips over consecutive nights. The number of clips recorded each night forms an arithmetic sequence. Over the first 10 nights the camera records a total of 730 clips. On the tenth night it records 100 clips.

(a) Show that the number of clips recorded on the first night is 46. **1**

(b) Find the common difference of the sequence. **1**

(c) Assuming the same pattern continues, find the first night by which the total number of clips recorded exceeds 1200. **2**

Sample answer

(a)

$S_{10}=\frac{10}{2}(a+100)=730 \Rightarrow a+100=146,\ a=46$

Marking criteria

Criteria	Marks
• Correct proof	1

(b)

$46+9d=100 \Rightarrow d=6$

Criteria	Marks
• Correct answer	1

(c)

$S_n=\frac{n}{2}[92+6(n-1)]=n(3n+43)>1200$

$3n^2+43n-1200>0 \Rightarrow n>14.1;\ S_{14}=1190,\ S_{15}=1320$, so night 15

Criteria	Marks
• Correct answer	2
• Correct inequality in n, or equivalent merit	1

Question 54 (3 marks) MODERATE

Adapted from Killara High School 2026 Trial HSC

In an arithmetic sequence, the sum of the first six terms is 78 and the sum of the first ten terms is 210. Find the first term and the common difference. **3**

Sample answer

$S_6 = 3(2a+5d) = 78 \Rightarrow 2a+5d = 26$

$S_{10} = 5(2a+9d) = 210 \Rightarrow 2a+9d = 42$

Subtracting: $4d = 16,\ d = 4;\ 2a = 6,\ a = 3$

Marking criteria

Criteria	Marks
• Correct values	3
• Forms two correct equations, or equivalent merit	2
• One correct equation, or equivalent merit	1

Question 55 (2 marks) MODERATE

Adapted from Merewether High School 2025 Trial HSC

The sum of the first 10 terms of an arithmetic sequence is 160. The sum of the first 20 terms is 520. What is the first term a of the sequence? **2**

Sample answer

$5(2a+9d) = 160 \Rightarrow 2a+9d = 32;\ 10(2a+19d) = 520 \Rightarrow 2a+19d = 52$

$10d = 20,\ d = 2;\ 2a = 14,\ a = 7$

Marking criteria

Criteria	Marks
• Correct answer	2
• Forms two correct equations, or equivalent merit	1

Question 56 (4 marks) MODERATE

Adapted from Normanhurst Boys High School 2026 Trial HSC

The sales team at Frontier phone company sells 12 000 phones in the first month of operation. They increase their sales by 800 phones each month on the preceding month's sales.

(a) Find the number of phones sold in the last month of the second year of operation. **1**

(b) Sampson, another phone company, commenced business at exactly the same time. Their sales team sells 5000 phones in their first month and increases sales by 1500 each month on the preceding month's sales. After how many months will Sampson's total sales exceed Frontier's total sales? **3**

Sample answer

(a)

$T_{24} = 12\,000 + 23 \times 800 = 30\,400$

(b)

Sampson: $\frac{n}{2}[10\,000 + 1500(n-1)]$; Frontier: $\frac{n}{2}[24\,000 + 800(n-1)]$

$8500 + 1500n > 23\,200 + 800n \Rightarrow 700n > 14\,700 \Rightarrow n > 21$

Sampson's total first exceeds Frontier's after 22 months

Marking criteria

(a)

Criteria	Marks
• Correct answer	1

(b)

Criteria	Marks
• Correct answer	3
• Correct inequality comparing the two sums, or equivalent merit	2
• Correct expression for one of the sums, or equivalent merit	1

Question 57 (5 marks) MODERATE

Adapted from North Sydney Boys High School 2025 Trial HSC

Consider the series $8 + 12x + 18x^2 + 27x^3 + \cdots$

(a) Show that the terms in this series form a geometric progression. **1**

(b) For what value(s) of x will this series have a limiting sum? **2**

(c) Find the limiting sum if $x = \frac{1}{4}$. **2**

Sample answer

(a)

$\frac{12x}{8} = \frac{18x^2}{12x} = \frac{27x^3}{18x^2} = \frac{3x}{2}$: constant ratio, so geometric

(b)

$\left|\frac{3x}{2}\right| < 1 \Rightarrow -\frac{2}{3} < x < \frac{2}{3}$

(c)

$r = \frac{3}{8}:\ S_\infty = \frac{8}{1-\frac{3}{8}} = \frac{64}{5}$

Marking criteria

(a)

Criteria	Marks
• Correct proof	1

(b)

Criteria	Marks
• Correct answer	2
• Uses $\|r\| < 1$, or equivalent merit	1

(c)

Criteria	Marks
• Correct answer	2
• Finds $r = 3/8$, or equivalent merit	1

Question 58 (4 marks) MODERATE

Adapted from North Sydney Girls High School 2025 Trial HSC

For a geometric series the third term is 20 and the sixth term is -160.

(a) Find the common ratio and the first term. **2**

(b) Find the sum of the terms from T_{10} to T_{20} inclusive. **2**

Sample answer

(a)

$r^3 = \frac{-160}{20} = -8 \Rightarrow r = -2;\ a = \frac{20}{4} = 5$

(b)

$S_{20} - S_9 = \frac{5(1-(-2)^{20})}{3} - \frac{5(1-(-2)^9)}{3}$

$= \frac{5}{3}[(1-1\,048\,576)-(1+512)] = -1\,748\,480$

Marking criteria

Criteria	Marks
• Correct values	2
• Finds r, or equivalent merit	1

Criteria	Marks
• Correct answer	2
• Uses S_{20} - S_9, or equivalent merit	1

Question 59 (5 marks) MODERATE

Adapted from North Sydney Girls High School 2026 Trial HSC

Given the arithmetic series $27 + x + y + z + 11 + \cdots$

(a) Find the values of x, y and z. **2**

(b) Find the number of terms if the series sums to 99. **3**

Sample answer

(a)

$27 + 4d = 11 \Rightarrow d = -4$

$x = 23,\ y = 19,\ z = 15$

(b)

$\frac{n}{2}[54 - 4(n-1)] = 99 \Rightarrow n(58 - 4n) = 198$

$2n^2 - 29n + 99 = 0 \Rightarrow (2n-11)(n-9) = 0$

n is a whole number, so $n = 9$

Marking criteria

Criteria	Marks
• Correct values	2
• Finds d = -4, or equivalent merit	1

Criteria	Marks
• Correct answer	3
• Correct quadratic in n, or equivalent merit	2
• Correct use of the sum formula, or equivalent merit	1

Question 60 (3 marks) MODERATE

Adapted from St George Girls High School 2025 Trial HSC

The sequence $6 - x,\ x - 8,\ x + 4$ is geometric. Determine the possible values of the common ratio. **3**

Sample answer

$(x-8)^2 = (6-x)(x+4) \Rightarrow x^2 - 16x + 64 = -x^2 + 2x + 24$

$2x^2 - 18x + 40 = 0 \Rightarrow (x-4)(x-5) = 0$

$x = 4$: $r = \frac{-4}{2} = -2$; $x = 5$: $r = \frac{-3}{1} = -3$

Marking criteria

Criteria	Marks
• Both ratios correct	3
• Finds the values of x, or equivalent merit	2
• Correct equation using equal ratios, or equivalent merit	1

Question 61 (2 marks) MODERATE

Adapted from Sydney Grammar School 2025 Trial HSC

Find the values of x such that the limiting sum $1 + (1-2x) + (1-2x)^2 + (1-2x)^3 + \cdots$ exists. **2**

Sample answer

$r = 1 - 2x$; limiting sum exists when $|1 - 2x| < 1$

$-1 < 1 - 2x < 1 \Rightarrow -2 < -2x < 0 \Rightarrow 0 < x < 1$

Marking criteria

Criteria	Marks
• Correct answer	2
• Uses $\lvert r\rvert < 1$ with $r = 1 - 2x$, or equivalent merit	1

Question 62 (3 marks) DIFFICULT

Adapted from Baulkham Hills High School 2025 Trial HSC

The common difference of an arithmetic sequence is -4. The sum of the first 30 terms is 23 940 and the sum of the last 30 terms is -1980. Find how many terms are in the sequence. **3**

Sample answer

$S_{30} = 15[2a + 29(-4)] = 23\,940 \Rightarrow 2a - 116 = 1596,\ a = 856$

Last 30 terms: $T_{n-29} + T_n = \frac{-1980}{15} = -132$: $[856 - 4(n-30)] + [856 - 4(n-1)] = -132$

$1712 - 4(2n - 31) = -132 \Rightarrow 2n - 31 = 461 \Rightarrow n = 246$

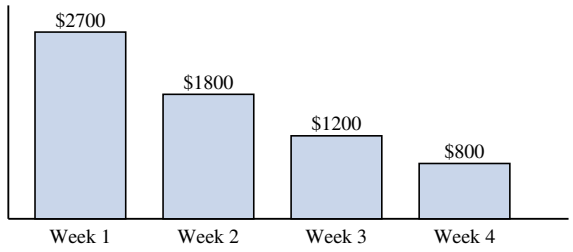
Marking criteria

Criteria	Marks
• Correct answer	3
• Correct equation for n using the last 30 terms, or equivalent merit	2
• Finds the first term, or equivalent merit	1

Question 63 (5 marks) MODERATE

Adapted from St George Girls High School 2026 / Roseville College 2026 Trial HSC

A charity records the donations received each week for the first four weeks of a fundraising drive, as shown. The director notices that the weekly donations follow a geometric sequence and assumes the pattern will continue.



(a) Find the common ratio. **1**

(b) How much would she predict to be received in Week 8 (to the nearest dollar)? **1**

(c) What total amount would be raised in the first thirteen weeks? **1**

(d) She claims the drive will never raise more than \$8100 in total. Explain how she can make this prediction and why she decides to launch a new drive after thirteen weeks. **2**

Sample answer

(a)

$r = \dfrac{1800}{2700} = \dfrac{2}{3}$

(b)

$T_8 = 2700\left(\dfrac{2}{3}\right)^7 = \$158$

(c)

$S_{13} = \dfrac{2700(1-(2/3)^{13})}{1-\frac{2}{3}} = \$8058.38$

(d)

$|r| < 1$ so the total approaches the limiting sum $S_\infty = \dfrac{2700}{1-\frac{2}{3}} = \$8100$

After 13 weeks \$8058 has been raised, only \$42 short of the \$8100 limit, so very little more can be collected and a new drive is needed

Marking criteria

Criteria	Marks
• Correct answer	1

Criteria	Marks
• Correct answer	1

Criteria	Marks
• Correct answer	1

Criteria	Marks
• Correct explanation referring to the limiting sum and comparing with S_{13}	2
• Finds the limiting sum of \$8100, or equivalent merit	1

Question 64 (4 marks) MODERATE

Adapted from Sydney Grammar School 2026 Trial HSC

A scientist prepares solutions whose hydrogen ion concentrations form a geometric sequence. Let H_n be the concentration of the nth solution. The first solution has concentration $H_1 = 10^{-2}$ mol/L and each subsequent solution has half the concentration of the previous one.

(a) Find the concentration of hydrogen ions in the nth solution. **1**

(b) The acidity of a solution is its pH value, $A(H) = -\log_{10} H$. Show that the acidity of the nth solution is $A_n = 2 + (n-1)\log_{10} 2$. **2**

(c) Determine the first solution in the sequence whose acidity exceeds 7. **1**

Sample answer

(a)

$H_n = 10^{-2}\left(\dfrac{1}{2}\right)^{n-1}$

(b)

$A_n = -\log_{10}\left[10^{-2}\left(\frac{1}{2}\right)^{n-1}\right] = -\left[-2 + (n-1)\log_{10}\frac{1}{2}\right]$

$= 2 - (n-1)(-\log_{10} 2) = 2 + (n-1)\log_{10} 2$

(an arithmetic sequence with $d = \log_{10} 2$)

(c)

$2 + (n-1)\log_{10} 2 > 7 \Rightarrow n - 1 > \dfrac{5}{\log_{10} 2} = 16.6$, so $n = 18$

Marking criteria

Criteria	Marks
• Correct answer	1

Criteria	Marks
• Correct proof	2
• Applies log laws to $-\log_{10} H_n$, or equivalent merit	1

Criteria	Marks
• Correct answer	1

Question 65 (3 marks) DIFFICULT

Adapted from Caringbah High School 2026 / Fort Street High School 2026 Trial HSC

Consider the function defined by the geometric series $f(x)=e^{-x}+e^{-2x}+e^{-3x}+\cdots$ for $x>0$.

(a) Explain why the limiting sum exists for $x>0$. **1**

(b) Show that $f(x)=\dfrac{1}{e^{x}-1}$. **2**

Sample answer

(a)

$r=e^{-x}$; for $x>0$, $0<e^{-x}<1$ so $|r|<1$ and the limiting sum exists

(b)

$f(x)=\dfrac{e^{-x}}{1-e^{-x}}$; multiply numerator and denominator by e^{x}:

$f(x)=\dfrac{1}{e^{x}-1}$

Marking criteria

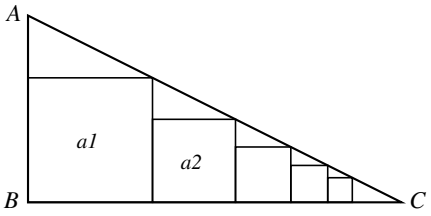
Criteria	Marks
• Correct explanation	1

Criteria	Marks
• Correct proof	2
• Correct use of the limiting sum formula, or equivalent merit	1

Question 66 (5 marks) DIFFICULT

Adapted from Fort Street High School 2025 Trial HSC

A sequence of squares is inscribed within a right-angled triangle ABC, where $AB=a$, $BC=2a$ and $\angle ABC=90^\circ$. Each square touches the hypotenuse AC, and the side lengths are $a_1,\ a_2,\ \ldots,\ a_n$.



(a) By considering similar triangles, show that $a_1=\dfrac{2}{3}a$. **1**

(b) Show that the areas of the squares form a geometric sequence. State the common ratio. **2**

(c) Find, in terms of a, the total sum of the areas of all the squares. **2**

Sample answer

(a)

The triangle above the first square has height $a-a_1$ and base a_1, similar to ABC: $\dfrac{a-a_1}{a_1}=\dfrac{a}{2a}=\dfrac{1}{2}$

$2a-2a_1=a_1 \Rightarrow a_1=\dfrac{2}{3}a$

(b)

The triangle to the right of each square is similar to ABC with height a_n (and base $2a_n$), so by (a) $a_{n+1}=\dfrac{2}{3}a_n$

Areas: $a_{n+1}^2=\dfrac{4}{9}a_n^2$, a geometric sequence with $r=\dfrac{4}{9}$

(c)

$S_\infty=\dfrac{a_1^2}{1-\frac{4}{9}}=\dfrac{\frac{4}{9}a^2}{\frac{5}{9}}=\dfrac{4a^2}{5}$

Marking criteria

Criteria	Marks
• Correct proof	1

Criteria	Marks
• Correct proof with ratio	2
• Shows $a_{n+1} = 2a_n/3$, or equivalent merit	1

Criteria	Marks
• Correct answer	2
• Correct use of the limiting sum formula, or equivalent merit	1

Question 67 (5 marks) DIFFICULT

Adapted from Girraween High School 2025 Trial HSC

Consider the geometric series $1-\tan^2\theta+\tan^4\theta-\cdots$

(a) When the limiting sum exists, find an expression for the limiting sum in simplest form. **3**

(b) For what values of θ in the interval $-\dfrac{\pi}{2}<\theta<\dfrac{\pi}{2}$ does the limiting sum of the series exist? **2**

Sample answer

(a)

$a=1,\ r=-\tan^2\theta$

$S_\infty=\dfrac{1}{1+\tan^2\theta}=\dfrac{1}{\sec^2\theta}=\cos^2\theta$

(b)

Need $|r|<1$: $\tan^2\theta<1 \Rightarrow -1<\tan\theta<1$

$-\dfrac{\pi}{4}<\theta<\dfrac{\pi}{4}$

Marking criteria

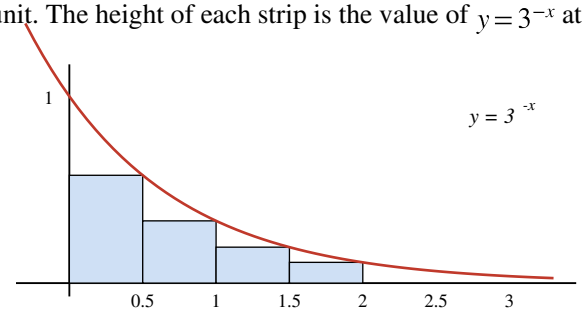
Criteria	Marks
• Correct simplified answer	3
• Correct unsimplified limiting sum, or equivalent merit	2
• Identifies $r = -\tan^2\theta$, or equivalent merit	1

Criteria	Marks
• Correct answer	2
• Uses $\tan^2\theta < 1$, or equivalent merit	1

Question 68 (3 marks) DIFFICULT

Adapted from Hornsby Girls High School 2026 Trial HSC

The diagram shows the graph of $y = 3^{-x}$ and the first four of an infinite number of rectangular strips of width $\frac{1}{2}$ unit. The height of each strip is the value of $y = 3^{-x}$ at the right-hand edge of the strip, i.e. at $x = \frac{1}{2},\ 1,\ \frac{3}{2},\ \ldots$



(a) The sum of the areas of the rectangles forms a geometric series. Show that the limiting sum of this series is $\frac{1}{2\sqrt{3}-2}$. **2**

(b) Express the limiting sum with a rational denominator. **1**

Sample answer

(a)

Areas: $\frac{1}{2}\cdot 3^{-1/2},\ \frac{1}{2}\cdot 3^{-1},\ \frac{1}{2}\cdot 3^{-3/2},\ \ldots$: $a = \frac{1}{2\sqrt{3}},\ r = \frac{1}{\sqrt{3}}$

$$S_\infty = \frac{\frac{1}{2\sqrt{3}}}{1-\frac{1}{\sqrt{3}}} = \frac{1}{2\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}-1} = \frac{1}{2\sqrt{3}-2}$$

(b)

$$\frac{1}{2(\sqrt{3}-1)} \times \frac{\sqrt{3}+1}{\sqrt{3}+1} = \frac{\sqrt{3}+1}{4}$$

Marking criteria

Criteria	Marks
• Correct proof	2
• Correct first term and ratio, or equivalent merit	1

Criteria	Marks
• Correct answer	1

Question 69 (5 marks) DIFFICULT

Adapted from Hunters Hill High School 2026 / Normanhurst Boys 2025 Trial HSC

ABC is a triangle. The numbers $1+\sin A,\ \frac{1}{2},\ 1-\sin A$ are the first three terms of a geometric sequence.

(a) Find the possible values of A. **3**

(b) Show that the limiting sum of the series $(1+\sin A)+\frac{1}{2}+(1-\sin A)+\cdots$ exists. **2**

Sample answer

(a)

$$\frac{1/2}{1+\sin A} = \frac{1-\sin A}{1/2} \Rightarrow \frac{1}{4} = 1-\sin^2 A$$

$$\sin^2 A = \frac{3}{4} \Rightarrow \sin A = \frac{\sqrt{3}}{2} \ (\sin A > 0 \text{ in a triangle})$$

$A = 60^\circ$ or 120°

(b)

$r = \frac{1/2}{1+\sin A} = \frac{1}{2+\sqrt{3}} \approx 0.27$ for both values of A

Since $|r| < 1$, the limiting sum exists

Marking criteria

Criteria	Marks
• Correct values	3
• Obtains $\sin^2 A = 3/4$, or equivalent merit	2
• Equates the common ratios, or equivalent merit	1

Criteria	Marks
• Correct proof	2
• Finds r, or equivalent merit	1

Question 70 (3 marks) DIFFICULT

Adapted from Manly Selective Campus 2026 Trial HSC

The limiting sum of a geometric series is 500, and the sum of the first three terms is 244. Find the sum of the first 10 terms, correct to two decimal places. **3**

Sample answer

$$S_3 = S_\infty(1-r^3) \Rightarrow 1-r^3 = \frac{244}{500} = 0.488 \Rightarrow r^3 = 0.512,\ r = 0.8$$

$$a = 500(1-0.8) = 100$$

$$S_{10} = \frac{100(1-0.8^{10})}{0.2} = 446.31$$

Marking criteria

Criteria	Marks
• Correct answer	3
• Finds r and a, or equivalent merit	2
• Obtains $r^3 = 0.512$, or equivalent merit	1

Question 71 (4 marks) DIFFICULT

Adapted from James Ruse Agricultural High School 2025 Trial HSC

Given the series $1\frac{1}{3}+2\frac{1}{9}+4\frac{1}{27}+8\frac{1}{81}+\ldots$, find an expression for

(a) the general term. **2**

(b) the sum of the first n terms. **2**

Sample answer

(a)

Whole parts $1,2,4,8,\ldots=2^{n-1}$; fractions $\frac{1}{3},\frac{1}{9},\ldots=\left(\frac{1}{3}\right)^n$

$T_n=2^{n-1}+\frac{1}{3^n}$

(b)

$S_n=\frac{1(2^n-1)}{2-1}+\frac{\frac{1}{3}(1-(1/3)^n)}{1-\frac{1}{3}}$

$=2^n-1+\frac{1}{2}\left(1-\frac{1}{3^n}\right)$

Marking criteria

Criteria	Marks
• Correct expression	2
• Recognises the sum of a geometric sequence and a second geometric sequence, or equivalent merit	1

Criteria	Marks
• Correct expression	2
• Correct sum of one of the geometric series, or equivalent merit	1

Question 72 (5 marks) DIFFICULT

Adapted from Knox Grammar School 2026 Trial HSC

Two arithmetic progressions, A and B, each have 100 terms denoted by a_i and b_i respectively, where $i=1,2,\ldots,100$. The common difference of A is d, a positive integer. The progressions satisfy $a_1=b_{100}=4$ and $b_1=a_{100}$.

(a) You are given that there is at least one value of i for which $b_i=10+a_i$. Show that, in this case, $i=\frac{101}{2}-\frac{5}{d}$. **3**

(b) Hence determine the value(s) of d for which $b_i=10+a_i$ is true. **2**

Sample answer

(a)

$a_i=4+(i-1)d$, $b_1=a_{100}=4+99d$, $b_{100}=4$, so B has common difference $\frac{4-(4+99d)}{99}=-d$

$b_i=4+99d-(i-1)d=4+(100-i)d$

$b_i=10+a_i$: $4+(100-i)d=14+(i-1)d \Rightarrow (101-2i)d=10 \Rightarrow i=\frac{101}{2}-\frac{5}{d}$

(b)

i must be a positive integer, so $\frac{5}{d}$ must be an odd multiple of $\frac{1}{2}$: $\frac{5}{d}=\frac{1}{2}$ or $\frac{5}{2}$

$d=10$ $(i=50)$ or $d=2$ $(i=48)$

Marking criteria

Criteria	Marks
• Correct proof	3
• Correct expression for b_i, or equivalent merit	2
• Finds the common difference of B is $-d$, or equivalent merit	1

Criteria	Marks
• Both values with justification	2
• One correct value, or equivalent merit	1

Question 73 (3 marks) DIFFICULT

Adapted from Normanhurst Boys High School 2025 Trial HSC

If $y^2=(2m-y)(2n-y)$, show that $\frac{1}{m}$, $\frac{1}{y}$, $\frac{1}{n}$ (in this order) form the terms of an arithmetic sequence. **3**

Sample answer

$y^2=4mn-2my-2ny+y^2 \Rightarrow 2my+2ny=4mn \Rightarrow my+ny=2mn$

Divide by mny: $\frac{1}{n}+\frac{1}{m}=\frac{2}{y}$, i.e. $\frac{1}{y}-\frac{1}{m}=\frac{1}{n}-\frac{1}{y}$

Equal differences, so $\frac{1}{m},\frac{1}{y},\frac{1}{n}$ is arithmetic

Marking criteria

Criteria	Marks
• Correct proof	3
• Obtains $my + ny = 2mn$, or equivalent merit	2
• Expands correctly, or equivalent merit	1

Question 74 (3 marks) DIFFICULT

Adapted from North Sydney Boys High School 2025 Trial HSC

Three numbers form a geometric sequence. If the first two terms remain unchanged and the third term is decreased by 32, they form an arithmetic sequence. Then, if the second term of this arithmetic sequence is decreased by 4, the sequence becomes geometric again. Find the third term of the original geometric sequence. **3**

Sample answer

Let the terms be a, ar, ar^2. AP: $2ar=a+ar^2-32$ ... (1)

GP: $(ar-4)^2=a(ar^2-32) \Rightarrow -8ar+16=-32a \Rightarrow ar=4a+2$ ... (2)

Sub (2) in (1): $8a+4=a+\frac{(4a+2)^2}{a}-32 \Rightarrow 9a^2-20a+4=0 \Rightarrow a=2$ or $\frac{2}{9}$

$a=2$: $r=5$, third term $=50$ (or $a=\frac{2}{9}$, $r=13$: third term $\frac{338}{9}$)

Marking criteria

Criteria	Marks
• Correct answer(s)	3
• Obtains a quadratic in a (or equivalent), or equivalent merit	2
• Forms two correct equations, or equivalent merit	1

Question 75 (4 marks) DIFFICULT

Adapted from North Sydney Girls High School 2025 Trial HSC

The numbers $4e^{-k},\ 5,\ 4e^{k}$ are consecutive terms in an arithmetic sequence.

(a) Show that $2e^{k}-5+2e^{-k}=0$. **1**

(b) Hence find in exact form any possible values of k. **3**

Sample answer

(a)

$5-4e^{-k}=4e^{k}-5 \Rightarrow 4e^{k}+4e^{-k}=10 \Rightarrow 2e^{k}-5+2e^{-k}=0$

(b)

Multiply by e^{k}: $2e^{2k}-5e^{k}+2=0 \Rightarrow (2e^{k}-1)(e^{k}-2)=0$

$e^{k}=2$ or $e^{k}=\frac{1}{2} \Rightarrow k=\ln 2$ or $k=-\ln 2$

Marking criteria

Criteria	Marks
• Correct proof	1

Criteria	Marks
• Correct values	3
• Solves the quadratic for e^{k}, or equivalent merit	2
• Obtains $2e^{2k} - 5e^{k} + 2 = 0$, or equivalent merit	1

Question 76 (4 marks) DIFFICULT

Adapted from North Sydney Girls High School 2026 Trial HSC

A gambler makes a profit each day but a loss each night. He reduces the amount he bets by a fixed percentage each session, so his profit/loss per session follows a geometric sequence: if his profit is x dollars on the first day, y dollars on the first night and z dollars on the second day, then $x+y+z+\ldots$ is a geometric series.

(a) Given the profit on the third day was \$27 and the loss on the fourth night was \$8, show that the ratio between successive sessions is $-\frac{2}{3}$ and find the profit on the first day. **2**

(b) How much profit or loss will the gambler eventually make? **2**

Sample answer

(a)

Day 3 is session 5, night 4 is session 8: $T_5=27,\ T_8=-8$

$r^3=\frac{-8}{27} \Rightarrow r=-\frac{2}{3}$

$ar^4=27 \Rightarrow a=27\times\frac{81}{16}=\$136.69$

(b)

$S_\infty=\frac{136.6875}{1+\frac{2}{3}}=\$82.01$ profit

Marking criteria

Criteria	Marks
• Correct proof and first-day profit	2
• Identifies $T_5 = 27$ and $T_8 = -8$, or equivalent merit	1

Criteria	Marks
• Correct answer	2
• Uses the limiting sum, or equivalent merit	1

Question 77 (4 marks) DIFFICULT

Adapted from Pymble Ladies' College 2025 Trial HSC

The first term of an arithmetic series is -20 and the common difference is 5.

(a) Find the sum of the first 20 terms of the progression. **1**

(b) It is given that the sum of the first $2k$ terms is ten times the sum of the first k terms. Find the value of k. **3**

Sample answer

(a)

$S_{20}=\frac{20}{2}[-40+19\times 5]=550$

(b)

$S_{2k}=k[-40+5(2k-1)]=k(10k-45);\ S_k=\frac{k}{2}[-40+5(k-1)]=\frac{k}{2}(5k-45)$

$k(10k-45)=5k(5k-45) \Rightarrow 10k-45=25k-225$

$15k=180 \Rightarrow k=12$

Marking criteria

Criteria	Marks
• Correct answer	1

Criteria	Marks
• Correct answer	3
• Correct equation in k, or equivalent merit	2
• Correct expressions for S_{2k} and S_k, or equivalent merit	1

Question 78 (4 marks) DIFFICULT

Adapted from St George Girls High School 2025 Trial HSC

Consider the infinite series $S = \sin^2 x\cos^2 x + 2\sin^2 x\cos^4 x + 4\sin^2 x\cos^6 x + \cdots$

(a) Given that the limiting sum of S exists, show that it is $\dfrac{\sin^2 x\cos^2 x}{\sin^2 x - \cos^2 x}$. **2**

(b) Hence show that $\dfrac{1}{\operatorname{cosec}^2 x(\tan^2 x - 1)}$ can also be expressed as the limiting sum of S. **2**

Sample answer

(a)

$a = \sin^2 x\cos^2 x,\ r = 2\cos^2 x$

$S_\infty = \dfrac{\sin^2 x\cos^2 x}{1 - 2\cos^2 x} = \dfrac{\sin^2 x\cos^2 x}{\sin^2 x - \cos^2 x}$ (as $1 = \sin^2 x + \cos^2 x$)

(b)

$\dfrac{1}{\operatorname{cosec}^2 x(\tan^2 x - 1)} = \dfrac{\sin^2 x}{\frac{\sin^2 x}{\cos^2 x} - 1} = \dfrac{\sin^2 x\cos^2 x}{\sin^2 x - \cos^2 x}$, the limiting sum of S

Marking criteria

Criteria	Marks
• Correct proof	2
• Identifies $r = 2\cos^2 x$, or equivalent merit	1

Criteria	Marks
• Correct proof	2
• Rewrites in terms of *sin x* and *cos x*, or equivalent merit	1

Question 79 (4 marks) DIFFICULT

Adapted from Sydney Grammar School 2026 Trial HSC

Consider the geometric series $\cos x + \cos^3 x + \cos^5 x + \cos^7 x + \cdots$, where $0 < x < \dfrac{\pi}{2}$.

(a) Show that the limiting sum of this series exists. **1**

(b) Hence solve $\cos x + \cos^3 x + \cos^5 x + \cos^7 x + \cdots = \dfrac{2}{3}$ for $0 < x < \dfrac{\pi}{2}$. **3**

Sample answer

(a)

$r = \cos^2 x$; for $0 < x < \dfrac{\pi}{2}$, $0 < \cos x < 1$ so $0 < r < 1$ and the limiting sum exists

(b)

$S_\infty = \dfrac{\cos x}{1 - \cos^2 x} = \dfrac{\cos x}{\sin^2 x} = \dfrac{2}{3}$

$3\cos x = 2(1 - \cos^2 x) \Rightarrow 2\cos^2 x + 3\cos x - 2 = 0 \Rightarrow (2\cos x - 1)(\cos x + 2) = 0$

$\cos x = \dfrac{1}{2}\ (\cos x \neq -2) \Rightarrow x = \dfrac{\pi}{3}$

Marking criteria

Criteria	Marks
• Correct justification	1

Criteria	Marks
• Correct solution	3
• Obtains a quadratic in *cos x*, or equivalent merit	2
• Correct expression for the limiting sum, or equivalent merit	1

Summary of Answers

Mathematics Advanced — Sequences and Series Practice Paper (2025–2026 trials)

Q	Lvl	Answer
1	E	C
2	E	C (73)
3	E	B (69)
4	E	D (44)
5	E	C (43)
6	E	C (495)
7	E	A ($S_n = n^2$)
8	E	D (4π)
9	E	C
10	E	B (−336½)
11	M	A (−18)
12	M	D (x = −1/3 or 3)
13	M	D (x = 6/5)
14	M	B ($2048x^{11}$)
15	M	C (d = 4)
16	M	D (19 954)
17	M	B
18	M	D ($p^2/(p - q)$)
19	D	C (r = 0.4)
20	E	384
21	D	A (−32 805)
22	D	D ($\log_{2e} 2$)
23	E	(a) r = 1/3 (b) 2/243 (c) 27
24	E	a = −23, d = 12
25	E	135 523
26	E	a = 1.5, r = 2
27	E	(a) $T_n = 58 - 7n$ (b) 15 (c) 30
28	E	x = −5
29	E	(a) d = 3, a = 2 (b) 8475
30	E	p = 320, q = 80
31	E	177 146
32	E	1125
33	E	(a) 256, 64, 16, 4 (b) 1024/3
34	E	a = 148/3
35	E	207
36	E	1460
37	E	(a) 2, 7, 12 (b) 6225
38	E	94
39	E	$r = 2e^{-2} \approx 0.27 < 1$, limiting sum exists
40	E	d = 3
41	E	513 591
42	E	152 (42nd term)
43	E	(a) 81 (b) 28 593
44	E	495
45	E	59 048
46	M	(a) 1 (b) 20 terms
47	M	15/14
48	M	(a) 17 hours (b) 61.1% remains
49	M	$T_n = 6n - 7$, d = 6
50	M	(a) 31 (b) 2/3 cm (c) 36 cm
51	M	643.5
52	M	(a) proof (b) 513 mm (c) constant difference 27 mm (d) 5670 mm
53	M	(a) proof (b) d = 6 (c) night 15
54	M	a = 3, d = 4
55	M	a = 7
56	M	(a) 30 400 (b) 22 months
57	M	(a) r = 3x/2 (b) −2/3 < x < 2/3 (c) 64/5
58	M	(a) r = −2, a = 5 (b) −1 748 480
59	M	(a) 23, 19, 15 (b) 9 terms
60	M	r = −2 or r = −3
61	M	0 < x < 1
62	D	246 terms
63	M	(a) 2/3 (b) $158 (c) $8058.38 (d) S_∞ = $8100
64	M	(a) $H_n = 10^{-2}(½)^{n-1}$ (b) proof (c) 18th solution
65	D	(a) $0 < e^{-x} < 1$ (b) proof
66	D	(a) proof (b) r = 4/9 (c) $4a^2/5$
67	D	(a) $\cos^2\theta$ (b) $-\pi/4 < \theta < \pi/4$
68	D	(a) proof (b) $(\sqrt{3} + 1)/4$
69	D	(a) A = 60° or 120° (b) $r = 1/(2 + \sqrt{3}) < 1$
70	D	446.31
71	D	(a) $2^{n-1} + 3^{-n}$ (b) $2^n - 1 + ½(1 - 3^{-n})$
72	D	(a) proof (b) d = 2 or 10
73	D	proof
74	D	50 (or 338/9)
75	D	(a) proof (b) k = ±ln 2
76	D	(a) $136.69 (b) $82.01 profit
77	D	(a) 550 (b) k = 12
78	D	proofs
79	D	(a) $0 < \cos^2 x < 1$ (b) $x = \pi/3$

Level key: E = Easy M = Moderate D = Difficult. Easy: 34 questions (67 marks) • Moderate: 26 questions (76 marks) • Difficult: 19 questions (65 marks) • **Total: 79 questions, 208 marks**.

Key formulae (NESA Reference Sheet — Sequences and series)

Arithmetic	$T_n = a + (n-1)d$	$S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}(a + l)$
Geometric	$T_n = ar^{n-1}$	$S_n = \frac{a(1-r^n)}{1-r} = \frac{a(r^n-1)}{r-1},\ r \neq 1$
Limiting sum	$S_\infty = \frac{a}{1-r},\ \lvert r\rvert < 1$	Exists only when the common ratio satisfies |r| < 1
Useful results	$T_n = S_n - S_{n-1}$	Three terms a, b, c: arithmetic if 2b = a + c; geometric if $b^2 = ac$